

Existence of maximal monotone extensions

Proof of Conjecture 00000008847

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4 October 2026

Theorem. Every set-valued monotone operator on a real Hilbert space admits a maximal monotone extension. Maximal elements in the inclusion order on monotone graphs are exactly maximal monotone operators. In fact, the proof works in every real inner-product space, without assuming completeness.

1. Definitions and scope

Let E be a real inner-product space. A set-valued operator is a map $A : E \rightarrow \mathcal{P}(E)$; empty values are allowed. Its graph is

$$\text{gra } A = \{(x, u) \in E \times E : u \in A(x)\}.$$

It is *monotone* when

$$\langle x - y, u - v \rangle \geq 0 \quad \text{whenever } u \in A(x), v \in A(y).$$

An operator B extends A when $A(x) \subseteq B(x)$ for all x , equivalently $\text{gra } A \subseteq \text{gra } B$. An operator is *maximal monotone* if it is monotone and has no proper monotone graph extension. This is maximality among monotone operators, not a claim that its values are maximal subsets of E .

These are the standard set-valued monotone-operator meanings of the conjecture. Both the extension assertion and the graph-order assertion are proved below. No continuity, nonempty-domain, or full-domain assumption is imposed. The proof is an application of the axiom of choice through Zorn's lemma and does not assert uniqueness or provide an algorithm for selecting an extension.

2. Monotonicity of chain unions

Call a subset $S \subseteq E \times E$ a monotone graph when every two of its points satisfy the displayed inner-product inequality. Let $\mathcal{C}$ be a family of monotone graphs linearly ordered by inclusion. Then $U = \bigcup_{S \in \mathcal{C}} S$ is a monotone graph.

Indeed, choose $(x, u), (y, v) \in U$. There are $S, T \in \mathcal{C}$ containing the first and second points, respectively. Since $\mathcal{C}$ is a chain, either $S \subseteq T$ or $T \subseteq S$. In the first case both points lie in T , and in the second both lie in S . Monotonicity of that graph gives $\langle x - y, u - v \rangle \geq 0$. This proves the claim. For the empty chain, its union is empty and monotonicity is vacuous.

3. Existence and maximality

Fix a monotone operator A and consider the set

$$\mathcal{M}_A = \{S \subseteq E \times E : \text{gra } A \subseteq S, S \text{ is a monotone graph}\},$$

ordered by inclusion. It is nonempty because $\text{gra } A \in \mathcal{M}_A$. Every nonempty chain in $\mathcal{M}_A$ has its union as an upper bound: the union contains $\text{gra } A$ and is monotone by Section 2. The empty chain has $\text{gra } A$ as an upper bound. Zorn's lemma therefore supplies a maximal element $M \in \mathcal{M}_A$.

Define $B : E \rightarrow \mathcal{P}(E)$ by

$$B(x) = \{u \in E : (x, u) \in M\}.$$

Then $\text{gra } B = M$, so B is monotone and extends A . If a monotone operator C extends B , then its graph also contains $\text{gra } A$, hence belongs to $\mathcal{M}_A$. Maximality of M gives $\text{gra } C = M = \text{gra } B$, and therefore $C = B$. Thus B is maximal monotone.

For the second assertion, the assignments

$$A \longmapsto \text{gra } A, \quad S \longmapsto [x \mapsto \{u : (x, u) \in S\}]$$

are inverse bijections between set-valued maps and subsets of $E \times E$. They preserve the extension/inclusion order and the defining monotonicity inequality. Hence a maximal element of the monotone-graph order is exactly a maximal monotone operator, and conversely. $\square$

4. Lean correspondence and verification

The accompanying Lean 4.19.0 project pins Mathlib to commit

`c44e0c8ee63ca166450922a373c7409c5d26b00b`.

The formal theorem is polymorphic in an arbitrary real inner-product space. The definitions `graph` and `ofGraph` implement the inverse bijections, with both inverse identities proved. The theorem `monotone_iff` exposes the ordinary pointwise inner-product inequality, and `graph_subset_iff` proves the order correspondence.

`monotone_sUnion` proves the chain-union lemma directly on graph points.

`exists_maximal_graph` applies Mathlib's `zorn_subset_nonempty`, whose initial graph parameter ensures that the maximal graph contains the prescribed seed. `maximal_graph_iff` proves the exact equivalence with `IsMaximalMonotone`; `exists_maximal_extension` transfers the graph result back to set-valued operators.

The final theorem `conjecture_00000008847` combines both original assertions. Its proof has no additional axioms, incomplete terms, or `native_decide`. Standard classical choice, as used in Zorn's lemma, is recorded by the axiom audit in `VERIFICATION.md`. No auxiliary numerical computation is involved.